

1. Consider digits dataset, Implement and Analyze Agglomerative Clustering, and Learn how to choose the number of clusters using a dendrogram. Visualize the Clusters and report the Silhouette Score to evaluate the quality of the clusters.

Clustering Target: *Group the images of digits into clusters representing each digit (0-9).*

Download the digits dataset using the code below

```
from sklearn import datasets  
data = datasets.load_digits()
```

2. Consider the wine dataset, Implement and Analyze Linear Discriminant Analysis (LDA), and classify the data. Use the previously implemented PCA and compare the results with Logistic Regression and Naive Bayes classifier. Evaluate and compare their performance using various metrics (For each classifier, report accuracy, precision, recall, F1 score, R^2 , and ROC-AUC score also visualize the confusion matrix for each model).

Download the wine dataset using the code below

```
from sklearn import datasets  
data = datasets.load_wine()
```